

Christopher Sparrow

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Available for Spring/Summer 2027 Co-op

EDUCATION

Northeastern University – Boston, MA

Expected May 2029

Bachelor of Science in Electrical Engineering

GPA: 3.84

Honors: Dean's List

Coursework: Embedded Design, Circuits and Signals, Differential Equations and Linear Algebra, Physics II

Affiliations: IEEE Student Member, MTT-S (Microwave Theory), AP-S (Antennas and Propagation), AESS (Aerospace)

TECHNICAL SKILLS

Languages: C/C++, MicroPython, MATLAB, Python

Hardware & Protocols: Raspberry Pi Pico, Arduino, PWM, I2C, UART, SPI

Design Tools: KiCad, LTSpice, PlatformIO, Quartus, SolidWorks, AutoCAD, 3D Printing, Laser Cutting

Lab Equipment: Oscilloscope, Digital Multimeter, DC Power Supply, Function Generator

PROJECTS

Ground Station Telemetry Receiver (solo) – AerospaceNU

September 2026 – Present

- Build a ground station that receives telemetry from team rocket avionics and relays decoded packets over Bluetooth to a laptop or phone for live tracking during flight
- Select the radio front end and antenna, lay out the board in KiCad, and write receive firmware

Custom GPS / Radio Avionics Printed Circuit Board (solo) – AerospaceNU

January 2026 – Present

- Designed a custom printed circuit board in KiCad that integrates GPS and radio modules for rocket avionics
- Develop embedded drivers so the board can parse GPS position fixes and drive the radio without a host computer
- Order components, hand solder the assembly, and bench test the board for signal integrity and power draw
- Target a flight-ready board that logs position and transmits telemetry to a ground station through launch and recovery

Automatic Toothpaste Dispenser – Northeastern University

January 2026 – April 2026

Electronics Lead, Team of 3

- Led electronics for an assistive dispenser that delivers dentist-recommended dose sizes for users with limited hand dexterity
- Programmed a Raspberry Pi Pico in MicroPython to drive a direct current gearmotor through a dual-channel motor driver, using pulse-width modulation and timed runs for adult and child dose settings
- Implemented 25 ms button debouncing and double-click detection so two inputs control dose size and motor direction

Avionics Flight Computer Firmware – AerospaceNU

September 2025 – December 2025

- Worked alongside mentors to engineer embedded I2C drivers that convert raw MS5607 barometer and ICM-20602 registers into SI units and altitude
- Applied low-pass filtering to altitude and velocity data, reducing noise and eliminating false state triggers
- Programmed 50 Hz telemetry logging to S25FL512 SPI flash and state detection logic for liftoff, apogee, and landing

ACTIVITIES & LEADERSHIP

Northeastern University Launch Initiative (AerospaceNU)

Fall 2025 – Present

Avionics Engineer

- Build the flight computers, telemetry, and control electronics that every subteam depends on
- Support projects such as automated air brakes and flight systems for a carbon fiber supersonic rocket

Northeastern Rugby Team (Division 1AA)

Fall 2025 – Present

First Team Player

- Coach new forwards through scrums, lineouts, and pod plays, including team-specific calls
- Compete on a squad that went 6-0 in the 2025 Liberty Rugby Conference to earn the top playoff seed

INTERESTS

Diving: Freediving, underwater filmmaking, spearfishing, Advanced Open Water and Enriched Air Nitrox scuba certified

Other: Solo travel, chess, reading historical nonfiction, Texas Hold'em poker